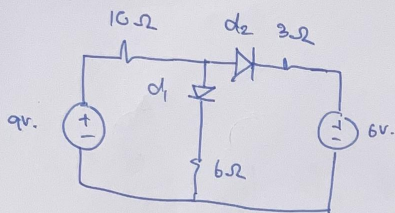


HW1

diodes are

Find out whether ~~diodes~~ ~~is~~ forward or reverse-biased

diode state	ideal	non-ideal
	ON	short
OFF	disconnected	disconnected



$$I_1$$

$$-9 + 10I_1 + 6I_1 - 6I_2 = 0$$

$$16I_1 - 6I_2 = 9$$

$$I_2$$

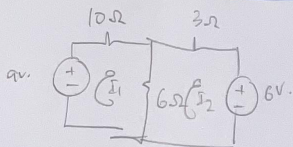
$$6I_2 - 6I_1 + 3I_2 + 6 = 0$$

$$-6I_1 + 9I_2 = -6$$

$$6I_1 - 9I_2 = 6$$

$$I_1 = \frac{5}{12}$$

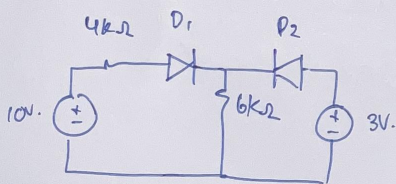
$$I_2 = -\frac{9}{18}$$



ideal

$$D_1 = I_1 - I_2 = \frac{5}{12} + \frac{9}{18} = \text{positive current} = \underline{\text{forward-biased}}$$

$$D_2 = I_2 = -\frac{9}{18} = \underline{\text{reverse-biased}}$$



$$I_1$$

$$-10 + 4kI_1 + 6kI_1 - 6kI_2 = 0$$

$$10kI_1 - 6kI_2 = 10$$

$$I_2$$

$$6kI_2 - 6kI_1 + 3 = 0$$

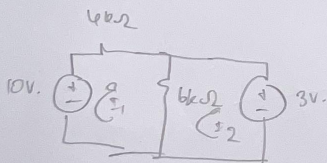
$$-6kI_1 + 6kI_2 = -3$$

$$6kI_1 - 6kI_2 = 3$$

$$I_1 = \frac{9}{4} \text{ mA}$$

$$I_2 = \frac{10}{8} \text{ mA}$$

assume all on



$$D_1 = I_1 = \frac{9}{4} \text{ mA} \quad \underline{\text{forward-biased}}$$

$$D_2 = -I_2 = -\frac{10}{8} \text{ mA} \quad \underline{\text{reverse-biased}}$$